DiffuMint-Stable-Diffusion-v1.5-LoRA-Fine-Tuning.ipynb

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Task

Fine-tune a Stable Diffusion v1.5 model on a custom local dataset using LoRA (Low-Rank Adaptation) optimized for a T4 GPU with 12GB of RAM. The task includes setting up the environment with `diffusers`, `transformers`, `accelerate`, and `bitsandbytes`, preparing images at 512x512 resolution, executing the training with memory-saving techniques like gradient checkpointing and fp16 precision, and finally saving and testing the LoRA weights by generating an inference sample.

[14] 6s

```
# 1. Install required libraries
!pip install -q diffusers transformers accelerate peft bitsandbytes xformers

# 2. Verify GPU
import torch
if torch.cuda.is_available():
    print(f"Using GPU: {torch.cuda.get_device_name(0)}")
else:
    print("No GPU found. Please enable T4 in Runtime settings.")
```

```
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.3/3.3 MB 53.4 MB/s eta 0:00:00
Using GPU: Tesla T4
```

Model Selection: Stable Diffusion v1.5 vs. SDXL

For a T4 GPU with **16GB VRAM** (standard in Colab), Stable Diffusion v1.5 is the preferred base model over SDXL for several reasons:

1. **Memory Footprint**: SD v1.5 has ~860M parameters, whereas SDXL has ~2.6B parameters. SDXL often requires significant

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Stable Diffusion v1.5 LoRA Fine-Tuning

1. Model Recommendation

Base Model: `runwayml/stable-diffusion-v1-5`
Reasoning: SD v1.5 is the gold standard for low-resource fine-tuning. It requires significantly less VRAM than SDXL, making it possible to train at 512x512 resolution on a free T4 GPU (12-16GB) without crashing.

2. Dataset Structure

Ensure your images are located in the following path:

```
/content/dataset/images/
└── img1.jpg
└── img2.png
...
```

3. Hyperparameters

- **Learning Rate:** 1e-4
- **Steps:** 500
- **LoRA Rank (r):** 8
- **LoRA Alpha:** 32
- **Precision:** fp16

Reasoning: Install the necessary libraries for LoRA training including diffusers, transformers, accelerate, bitsandbytes, and peft.

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[1]

```python
import os
from PIL import Image
import numpy as np

# 1. Create directory structure
os.makedirs('dataset/images', exist_ok=True)

# 2. Generate placeholder images for demonstration
print("Generating placeholder images...")
for i in range(3):
    # Create random colored images of different initial sizes
    random_image = np.random.randint(0, 256, (256 * (i+1), 256 * (i+1), 3), dtype=np.uint8)
    img = Image.fromarray(random_image)
    img.save(f'dataset/images/sample_{i}.jpg')

# 3. Define resizing function
def prepare_images(input_dir, size=(512, 512)):
    valid_extensions = ('.jpg', '.jpeg', '.png')
    image_files = [f for f in os.listdir(input_dir) if f.lower().endswith(valid_extensions)]

    for filename in image_files:
        img_path = os.path.join(input_dir, filename)
        with Image.open(img_path) as img:
            # Resize using Lanczos for high-quality downsampling
            img_resized = img.resize(size, Image.Resampling.LANCZOS)
            img_resized.save(img_path)
            print(f"Processed {filename}: {img.size} -> {img_resized.size}")

# 4. Execute resizing
prepare_images('dataset/images')

# 5. Verify results
final_files = os.listdir('dataset/images')
print(f"\nVerification: Found {len(final_files)} images in 'dataset/images'.")
for f in final_files:
    with Image.open(os.path.join('dataset/images', f)) as img:
        print(f"- {f}: {img.size} {img.format}")
```

```
Generating placeholder images...
Processed sample_2.jpg: (768, 768) -> (512, 512)
Processed sample_1.jpg: (512, 512) -> (512, 512)
Processed sample_0.jpg: (256, 256) -> (512, 512)

Verification: Found 3 images in 'dataset/images'.
- sample_2.jpg: (512, 512) JPEG
- sample_1.jpg: (512, 512) JPEG
```

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Fine-Tuning Configuration and LoRA Setup

Subtask:

Configure the training parameters and initialize the LoRA weights to minimize trainable parameters while optimizing for a 12GB-16GB RAM environment.

Fine-Tuning Configuration

To optimize training for a T4 GPU (12-16GB VRAM), we implement several memory-saving techniques:

1. **LoRA (Low-Rank Adaptation)**: Only training a small subset of parameters (adapters) instead of the full model.
2. **Mixed Precision (fp16)**: Reduces memory usage and speeds up computations.
3. **8-bit Adam Optimizer**: Significantly lowers the memory footprint of optimizer states.
4. **Gradient Checkpointing & Accumulation**: Allows for larger effective batch sizes by trading compute for memory.

Reasoning: I will initialize the training configuration, hyperparameters, and apply the LoRA configuration to the UNet model using the PEFT library as specified in the instructions.

[3]
✓ 2m

```
from diffusers import DDPMScheduler, UNet2DConditionModel
from transformers import CLIPTextModel, CLIPTokenizer
from peft import LoraConfig, get_peft_model

# 1. Define Training Configuration Constants
MODEL_NAME = "runwayml/stable-diffusion-v1-5"
OUTPUT_DIR = "sd-lora-output"
RESOLUTION = 512

# 2. Set Hyperparameters Optimized for T4 GPU (12GB-16GB RAM)
train_batch_size = 1
gradient_accumulation_steps = 4
learning_rate = 1e-4
```

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2m

```
unet.print_trainable_parameters()
```

```
Flax classes are deprecated and will be removed in Diffusers v1.0.0. We recommend migrating to PyTorch classes or pinning your version of Diffusers.
Flax classes are deprecated and will be removed in Diffusers v1.0.0. We recommend migrating to PyTorch classes or pinning your version of Diffusers.
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your Google Colab and restart your sess
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
tokenizer_config.json: 100%  806/806 [00:00<00:00, 21.7kB/s]
vocab.json:  1.06M/? [00:00<00:00, 12.1MB/s]
merges.txt:  525k/? [00:00<00:00, 7.80MB/s]
special_tokens_map.json: 100%  472/472 [00:00<00:00, 14.6kB/s]
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_validators.py:202: UserWarning: The `local_dir_use_symlinks` argument is deprecated and ignored in `hf_hub_download`.
  warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
scheduler_config.json: 100%  308/308 [00:00<00:00, 20.3kB/s]
config.json: 100%  617/617 [00:00<00:00, 19.9kB/s]
text_encoder/model.safetensors: 100%  492M/492M [00:05<00:00, 386MB/s]
Loading weights: 100%  196/196 [00:00<00:00, 404.55it/s, Materializing param=text_model.final_layer_norm.weight]
CLIPTextModel LOAD REPORT from: runwayml/stable-diffusion-v1-5
Key                              | Status     |  |
---------------------------------+------------+--+-
text_model.embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
config.json: 100%  743/743 [00:00<00:00, 26.6kB/s]
unet/diffusion_pytorch_model.safetensors: 100%  3.44G/3.44G [01:07<00:00, 43.6MB/s]
Model loaded. Trainable parameters:
trainable params: 1,594,368 || all params: 861,115,332 || trainable%: 0.1852
```

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[5] 13m

```
            accelerator.backward(loss)
            optimizer.step()
            optimizer.zero_grad()

        if accelerator.sync_gradients:
            global_step += 1
            progress_bar.update(1)
            if global_step % 50 == 0:
                vram_used = torch.cuda.memory_reserved() / 1024**3
                print(f" Step {global_step} | Loss: {loss.item():.4f} | VRAM Reserved: {vram_used:.2f} GB")

        if global_step >= max_train_steps:
            break

print("Training completed successfully.")
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_validators.py:202: UserWarning: The `local_dir_use_symlinks` argument is deprecated and ignored in `hf_hub_download`.
  warnings.warn(
config.json: 100%|| 547/547 [00:00<00:00, 21.6kB/s]
vae/diffusion_pytorch_model.safetensors: 100%|| 335M/335M [00:03<00:00, 139MB/s]
Starting training loop...
100%|| 500/500 [13:45<00:00, 1.68s/it]
 Step 50 | Loss: 0.1797 | VRAM Reserved: 5.18 GB
 Step 100 | Loss: 0.0571 | VRAM Reserved: 5.18 GB
 Step 150 | Loss: 0.0107 | VRAM Reserved: 5.18 GB
 Step 200 | Loss: 0.0022 | VRAM Reserved: 5.18 GB
 Step 250 | Loss: 0.0013 | VRAM Reserved: 5.18 GB
 Step 300 | Loss: 0.5157 | VRAM Reserved: 5.18 GB
 Step 350 | Loss: 0.0196 | VRAM Reserved: 5.18 GB
 Step 400 | Loss: 0.5393 | VRAM Reserved: 5.18 GB
 Step 450 | Loss: 0.0041 | VRAM Reserved: 5.18 GB
 Step 500 | Loss: 0.0128 | VRAM Reserved: 5.18 GB
Training completed successfully.
```

[12] 33s

```
import os
from diffusers import StableDiffusionPipeline
```

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```
Loading weights: 100%|▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰| 196/196 [00:02<00:00, 121.21it/s, Materializing param=text_model.final_layer_norm.weight]
CLIPTextModel LOAD REPORT from: /root/.cache/huggingface/hub/models--runwayml--stable-diffusion-v1-5/snapshots/451f4fe16113bff5a5d2269ed5ad43b0592e9a14/text_encoder
Key                                | Status     |  |
-----------------------------------+------------+--+-
text_model.embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
You have disabled the safety checker for <class 'diffusers.pipelines.stable_diffusion.pipeline_stable_diffusion.StableDiffusionPipeline'> by passing `safety_checker=None`. Ensure t
No LoRA keys associated to UNet2DConditionModel found with the prefix='unet'. This is safe to ignore if LoRA state dict didn't originally have any UNet2DConditionModel related para
No LoRA keys associated to CLIPTextModel found with the prefix='text_encoder'. This is safe to ignore if LoRA state dict didn't originally have any CLIPTextModel related params. Yo
Generating: a photo of an object
100%|▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰| 30/30 [00:06<00:00,  4.87it/s]
```

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```
You have disabled the safety checker for <class 'diffusers.pipelines.stable_diffusion.pipeline_stable_diffusion.StableDiffusionPipeline'> by passing `safety_checker=None`. Ensure
No LoRA keys associated to UNet2DConditionModel found with the prefix='unet'. This is safe to ignore if LoRA state dict didn't originally have any UNet2DConditionModel related pa
No LoRA keys associated to CLIPTextModel found with the prefix='text_encoder'. This is safe to ignore if LoRA state dict didn't originally have any CLIPTextModel related params.
Colab notebook detected. This cell will run indefinitely so that you can see errors and logs. To turn off, set debug=False in launch().
* Running on public URL: https://cb459b6068e2f0b2bc.gradio.live

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run `gradio deploy` from the terminal in the working directory to deploy to Hugging Face Spaces (h
```

Prompt

a photo of dog

Negative Prompt

low quality, blurry, distorted

Steps 30

1 50

Guidance Scale 7.5

1 15

Clear Submit

Generated Result

```
100% 30/30 [00:08<00:00, 3.85it/s]
/usr/local/lib/python3.12/dist-packages/diffusers/configuration_utils.py:141: FutureWarning: Accessing config attribute `__iter__` directly via 'DDPMScheduler' object attribute i
  deprecate("direct config name access", "1.0.0", deprecation_message, standard_warn=False)
/usr/local/lib/python3.12/dist-packages/diffusers/configuration_utils.py:141: FutureWarning: Accessing config attribute `__iter__` directly via 'StableDiffusionPipeline' object a
```

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